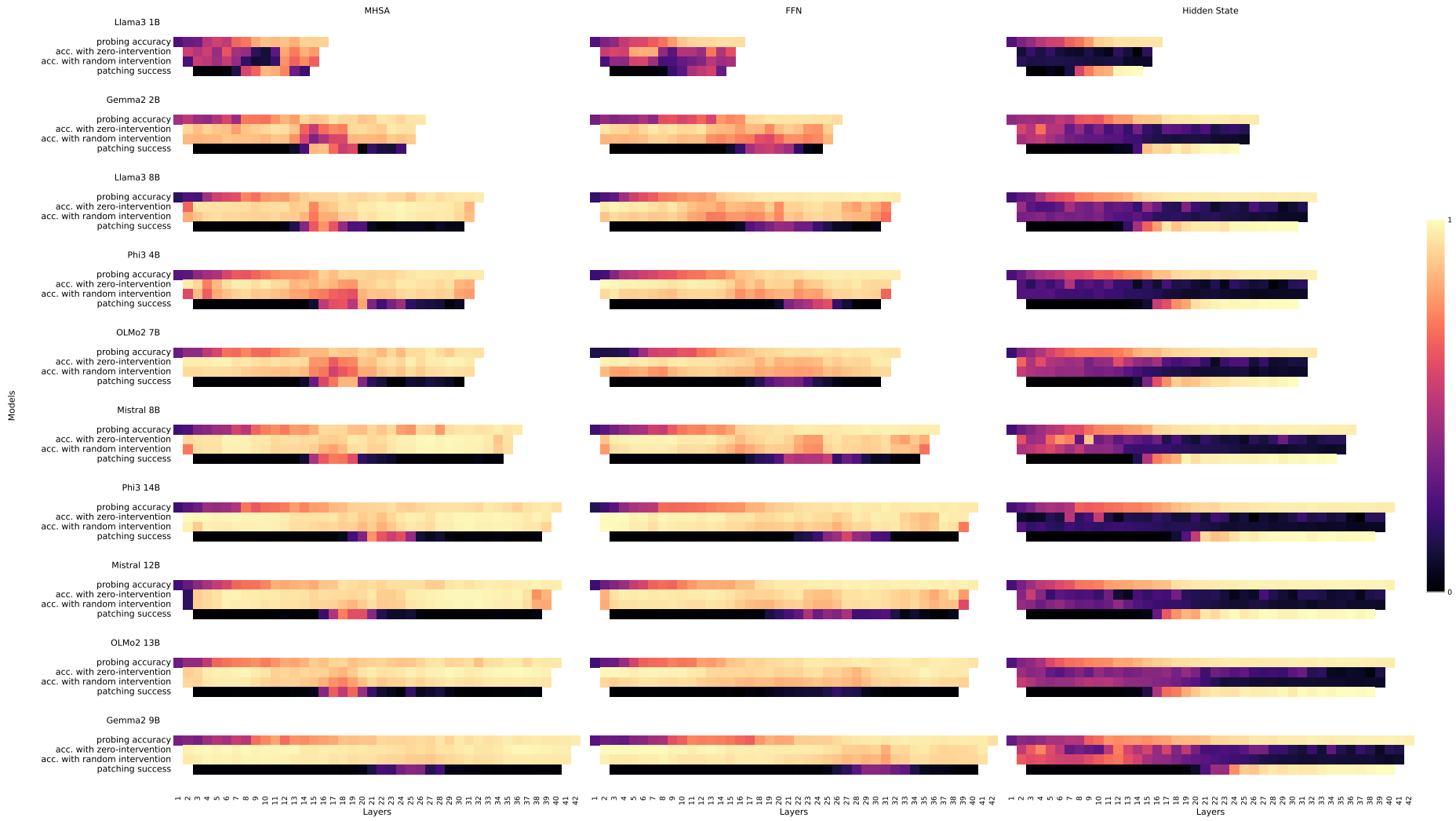

Figure 12: Comparison of probing, zero-activation and random-activation interventions, and activation patching on MSHA, FFN, and hidden state units across all layers of Llama 3.2 1B. The probe heatmap shows accuracy on the holdout set, zero/random activation interventions measure the model accuracy after disrupting causal pathways, and patching heatmaps indicate how effectively outputs transfer from source to target examples. The span sizes of 3 and 5 are used for the presented knockout interventions and patching experiments. This means that we intervene simultaneously on three/five consecutive layers, with the center being the indicated layer.

C Case Studies on Llama 3.2 1B

In this section, we focus on the Llama 3.2 1B language model and investigate the validity of our findings from multiple aspects. In Section C.1, we provide detailed results on emotion probing, activation patching and knockout interventions. In Section C.3 we extend our studies to include token dimension. Finally, we show that our results are robust to prompt design by conducting experiments using several hand-designed prompts in Section C.5.

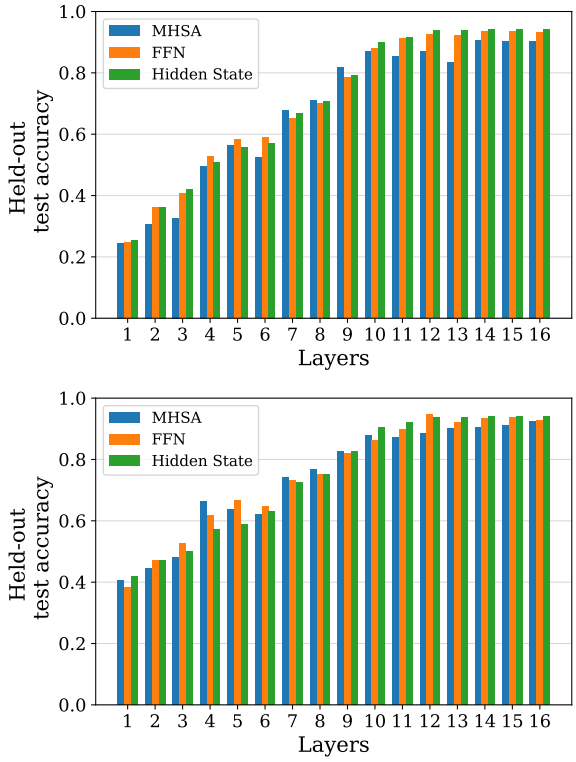

Figure 13: Probing test accuracy on last token of llama 3.2 1B for all layers. **Top** linear probe, **bottom** non-linear probe. There is a noticeable increase in probe performance in early layers when using a simple non-linear probe.

C.1 Details on Probing and Causal-Intervention

Here, we report the results of probing and causal intervention experiments on the Llama 3.2 1B model, focusing on the last token index across all layers, along with a simplified illustration of the findings. Figure 13 top demonstrates linear probing results.

It is noteworthy that we use linear probes to detect and extract emotion vectors. Low probe accuracy on earlier layers does not mean that there is no emotion signal at early layers but rather suggests that the signal is not *linearly* identifiable. In fact, Figure 13 bottom shows that when using a non-linear probe, i.e. a simple neural network with one hidden layer, the probe on earlier layers boosts considerably. However, both linear and non-linear probes peak around layer 10. After this layer, the model has finalized its output label decision and no major change happens.

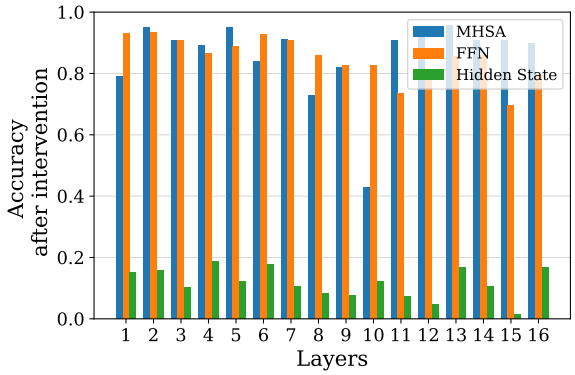

Figure 14: Zero-Intervention accuracy with span 1 on the last token index of Llama 3.2 1B across all layers. There is a clear drop in accuracy when MHSA activations in layer 10 are knocked out.

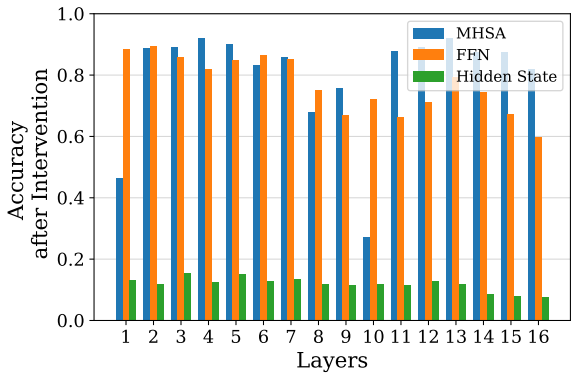

Figure 15: Random-Intervention accuracy with span 1 on the last token index of Llama 3.2 1B across all layers. A noticeable drop in accuracy is observed when MHSA activations in layer 10 are knocked out.

Intervention experiments further support this observation. Figures 14 and 15 show the effects of zero and random interventions with a span of 1, revealing a clear drop in accuracy when knocking out activations at layer 10. Finally, Figure 16 provides a detailed visualization of the patching experiment. The left-most plot highlights that the most successful emotion transfers occur at layer 10, which also exhibits the lowest number of unchanged labels. Notably, while some labels shifted to semantically similar emotions, they did not exactly match the

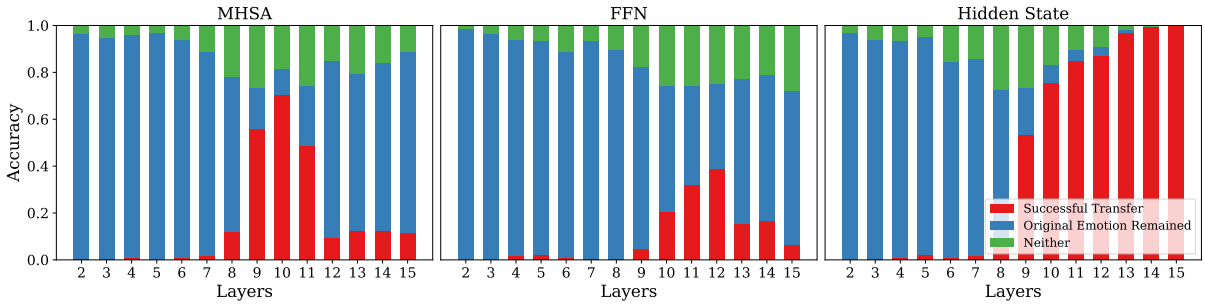

Figure 16: Activation patching results for Llama 3.2 1B across different layers (FFN, MHSA, and hidden state units) with Span = 3, evaluated over 200 source-target pairs. Blue indicates unsuccessful patching where the original label remained unchanged, red represents successful patching, and green denotes cases where the label changed but did not match the exact expected target.

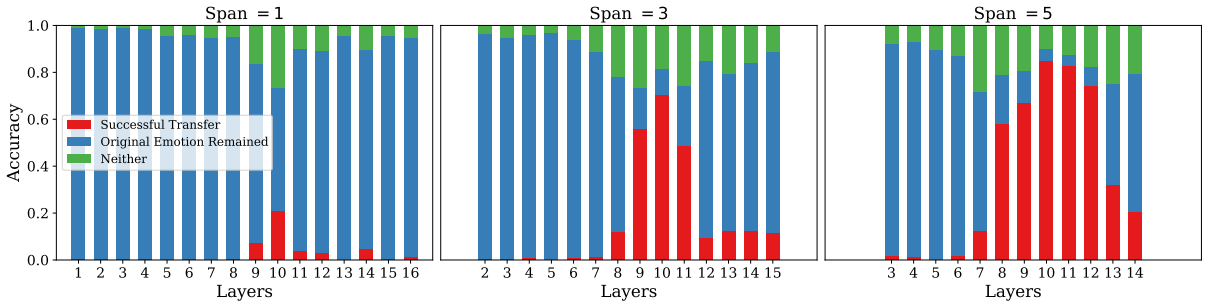

Figure 17: Effect of span size on Llama 3.2 1B activation patching at MHSA across different layers, evaluated over 200 source-target pairs.

target label and were, therefore, not counted as successful patches.

C.2 Effect of Span Size on Activation Patching

To examine the effect of span size on patching success, we repeated the experiment with three span sizes—1, 3, and 5 layers—on Llama 3.2 1B. Figure 17 presents the results, showing a clear increase in patching effectiveness as the span size increases. Notably, layer 10 continues to exhibit peak patching performance, reinforcing the idea of emotion-related functional localization in that layer.

C.3 Investigating Token Dimension

Throughout the paper, we extracted and analyzed the last token index activations $\mathbf{x}^{(l)}$—selected from one of the $\{\mathbf{a}^{(l)}, \mathbf{m}^{(l)}, \mathbf{h}^{(l)}\}$—across different layers of all tested models. We hypothesized that the strongest emotion signals appear at the last token, as it directly influences the model's next-word prediction in a causal language modeling setup. Earlier work demonstrates the presence of strong causal states immediately before the prediction, as well as their emergence at the final token of a noun phrase (Meng et al., 2022). Therefore, we suspect whether the last token of the query part in the prompt contains information more significant than the final token of the whole prompt.

To evaluate this hypothesis, we repeated the experiment on the last five tokens for Llama 3.2 1B extending the analysis to also include the final tokens in the query context (colored purple in Figure 4-Top). In Figure 18, we observe a consistent increase in signal strength from earlier to later tokens, reinforcing the focus on the last token as the primary contributor to output generation, but no clear importance on the last token of the query part.

Similarly, we conduct zero-activation intervention, random-activation intervention, and activation patching on Llama 3.2 1B's last five tokens across all layers. Figures 19, 20, and 21 confirm our hypothesis, suggesting that the last token's MSHA units in mid-layers, particularly $l = 10$, are critical for processing emotional content, while other token positions exhibit no localization.

Furthermore, the first row of Figures 19 and 20 illustrates the effect of zero or random activation interventions applied to all token positions in a layer